

PULSE

Solana Blitz v6 · Master Product, Hackathon, Infrastructure & Expert Frontend Prompt Pack

Purpose: hand this single PDF to a specialist frontend / mobile UI AI. It must contain every fact needed to ship a competition-grade mobile-first product without re-asking product questions. Start with the landing page only. Refine until perfect before deploy.

Document version	1.0 — 2026-07-11
Product codename	PULSE
Hackathon	Solana Blitz v6 (MagicBlock)
Theme	Mobile
Primary prize path	1st place \$500 USDC reserved for best mobile project
Repo	https://github.com/daraijaola/pulse
Live preview (VM)	https://agentr.online/sites/pulse/
Local path	C:\Users\HP\pulse
Stack (FE)	Vite + React + TypeScript · mobile-first PWA
Theme reference site	https://overflow.sui.io/ (visual system only — do NOT brand as Overflow)

Critical process rule: Do NOT one-shot and ship. Design → self-critique → refine → hard-check on phone viewport → only then deploy to VM. If it is not perfect, redo.

1. Hackathon details (complete)

1.1 Event identity

Name	Solana Blitz v6
Host / presenter	MagicBlock Team
Series	Weekend hackathon series for real-time apps on Solana
Format	Fully virtual
Category	Crypto / Solana / mobile-first
Registration Luma	https://luma.com/zsdnlbnt
Hackathon hub	https://hackathon.magicblock.app/
MagicBlock site	https://www.magicblock.xyz/
Docs	https://docs.magicblock.gg/
Calendar	https://luma.com/magicblock-events
Telegram support	https://t.me/+oLOcE79hoqo3OWJi
WorkAdventure cowork	https://play.workadventu.re/@/blitz/blitz/small-office (or MagicBlock office link on Luma)
RFP ideas (optional)	Notion RFP board linked from hackathon.magicblock.app

1.2 Dates & clock

Build window	Friday 10 July 2026 → Sunday 12 July 2026
Luma start_at (UTC)	2026-07-10T15:00:00.000Z

Luma end_at (UTC)	2026-07-12T15:00:00.000Z
Event timezone label	Asia/Hong_Kong on Luma event object
Hub copy (approx)	Monthly · Friday to Sunday · Asia/Singapore (series framing)
Hard deadline	Treat Sunday ~15:00 UTC as death; submit earlier. MagicBlock X also cited Sunday ~2PM UTC for prior messaging — verify submission Luma event.
Submission channel	Luma event: “Submission: Solana Blitz v6” on MagicBlock calendar
Submit package	GitHub repo + short demo video and/or live link

1.3 Prizes (\$1,000 USDC pool)

1st	\$500 USDC — Luma: best mobile build
2nd	\$250 USDC
3rd	\$150 USDC
Wizardio’s Choice	\$100 USDC

Eligibility (facts from Luma + hub): Projects should integrate MagicBlock **Ephemeral Rollups (ER)** and/or **Private ERs**. First prize is framed around the best mobile project. Judging: creativity, technical depth, how compellingly the project showcases what is possible on Solana with MagicBlock.

1.4 Participation logistics

- Register on Luma (wallet / Solana address verification may be required).
- Registration questions historically include Telegram username, GitHub username, optional X handle.
- Solo allowed; teams up to ~4 mentioned in MagicBlock communications.
- Partners called out for v6 mobile edition: Solana Mobile + Blueshift (orientation resources).
- Mobile docs: <https://docs.solanamobile.com/> · Blueshift Mobile Mastery path on learn.blueshift.gg
- Luma “Going” count is RSVPs/tickets — not guaranteed submissions. Past Blitz editions shipped ~30–40 projects.
- Post-Blitz path: MagicBlock Forge (rolling program) for strong builders.

1.5 MagicBlock product surface (what judges reward)

- Ephemeral Rollup (ER): high-frequency / real-time execution, delegate → act on ER → commit / undelegate to base layer.
- Private Ephemeral Rollup (PER): privacy / TEE path (optional for us unless stretch).
- VRF: verifiable onchain randomness (official roll-dice examples).
- Magic Router + regional ER RPCs (devnet free endpoints documented).
- Official examples: <https://github.com/magicblock-labs/magicblock-engine-examples> (counter, roll-dice, session keys, etc.).
- Dev skill (optional for agents): [magicblock-dev-skill](#) on GitHub.

1.6 Past winners — patterns to beat (facts)

v5 (Jun 12–14 2026, trading, ~34 apps): 1st Ghost Stops — ER recomputes triggers every tick; 2nd Eclipse — private state in ER; Wizardio EPOCH — market as ER session. **v3:** privRoll private payroll via PER/Payment API. Win pattern: one sharp mechanic + MagicBlock primitive is load-bearing + live demo + repo.

1.7 Official dev endpoints (devnet)

Magic Router: <https://devnet-router.magicblock.app>
ER (Asia example): <https://devnet-as.magicblock.app> · <wss://devnet-as.magicblock.app>
Solana Devnet: <https://api.devnet.solana.com>
Asia ER validator pubkey: MAS1Dt9qreoRMQ14YQuhg8UTZMMzDdKhmkZMECCzk57

2. Product: PULSE (complete build brief)

2.1 One-liner & thesis

PULSE is a mobile-first real-time reaction battle game on Solana. Players wait for a fair pulse signal, tap as fast as possible, and win the round. Room state and taps run on a **MagicBlock Ephemeral Rollup**. The GO timing uses **MagicBlock VRF** (not browser Math.random). The result commits to the Solana base layer.

Judge sentence: "Wait for a fair pulse → tap first → win — live on ER, settled on Solana — built for the phone."

2.2 Core loop (canonical)

- Open on phone browser / PWA.
- Create room OR join with 4-character code.
- Lobby: show code, you vs opponent (or Ghost bot for solo demo).
- Start round → delegate room account to Ephemeral Rollup.
- Request VRF → fair pulse / GO moment.
- WAIT phase (no early valid tap) → GO phase → player TAP (ER high-frequency tx).
- Score: reaction time / first valid tap (ghost opponent for solo).
- Commit / undelegate → winner + scores on base layer.
- Result screen → play again or home.

2.3 Frontend screens (build order)

Work **lead-source by lead-source**: finish one screen to perfection before the next. Current priority: **Landing first**.

Screen A — Landing (PRIORITY 1)

- Purpose: sell the product in <5 seconds on a phone; create desire to play.
- Must feel competition-winning UI: simple AND fire — not empty, not cluttered, not rainbow spam.
- Must use Overflow visual system (tokens, type, hard chrome) WITHOUT saying "Overflow" on screen.
- Must communicate the game (wait → pulse → tap → win) via designed motion / hero object — not a cheap mock dashboard.
- Primary CTA → Enter arena / Play (goes to Enter flow).
- Only product facts that matter (prefer 2 strong facts): fair pulse (VRF); live ER → Solana settle.
- No game dock chrome on landing. No fake score spam. No theme attribution footer.
- Logo: temporary wordmark OK; final logo later.
- Success bar: human team lead says "perfect" after phone review. Iterate until then.

Screen B — Enter (create / join)

- Create room (generates code) → Lobby.
- Join with code input (uppercase mono, 3–6 chars) → Lobby.
- Back to landing.
- Overflow buttons (navy primary + blue icon strip + diagonal arrow).

Screen C — Lobby

- Huge readable room code (camera / share).
- You vs Opponent/Ghost status cards.
- Start round CTA; Leave.
- Optional one-line MagicBlock flow hint (delegate next).
- App chrome: top brand bar + bottom dock OK here (not on landing).

Screen D — Arena

- Status line phases: delegating → VRF → waiting → GO → tapped → settling.
- Large thumb TAP target centered; designed pulse/signal object.
- Live scores you vs opp.
- Must work for phone recording (demo video).

- ER-visible UX (labels / phase badge) without clutter.

Screen E — Result

- Win / lose state, reaction ms, scores.
- Play again / Home.
- Optional short “judge line” about MagicBlock (not theme plagiarism).

2.4 In scope / out of scope

IN (MVP)

- Mobile-first PWA web app (Vite + React + TS).
- Wallet connect (Phantom / mobile wallet browser) when wiring chain.
- Onchain room + ER + VRF + settle (devnet first).
- Ghost opponent for solo demo.
- Deploy on team VM under /sites/pulse/ with basePath /sites/pulse/.
- README / demo script for judges.

OUT

- Native Seeker/Android as MVP requirement (PWA first).
- Token economy, betting, NFT mints, leaderboards backend, chat.
- Multi-mode game catalog.
- Claiming we reimplemented memory/randomness without MagicBlock.

2.5 On-chain model (target)

Room PDA fields (target): host, players[2], status, vrf_seed / pulse timing, winner, scores[]. Instructions (target): initialize_room, join_room, delegate_room, request_pulse (VRF), callback_pulse, tap (ER), commit_result / undelegate. Fork patterns from magicblock-engine-examples: counter (delegate/commit) + roll-dice (VRF).

2.6 Design system — Overflow reference (FACTS)

Study and match the visual language of <https://overflow.sui.io/> (Webflow CSS tokens). This is a design reference only. Do not put Overflow logos, “Overflow system”, or Sui branding in the product.

Color tokens (verbatim from Overflow shared CSS)

```
--color--blue-900: #000f1d (primary ink / navy)
--color--blue-700: #4da2ff
--color--blue-500: deepskyblue
--color--blue-400: #6de6f8
--color--light: #f7f7f7 (page bg)
--color--gray-200: #dfdfdf · gray-400: #bbbec0 · gray-700: #61686d · gray-800: #31404e
--color--orange-700: #ff7a00 · orange-200: #f2eee4 (use sparingly)
--color--pink-700: #ff6ada · pink-400: #e9ccff · pink-200: #dbcdeb (use sparingly)
--color--purple-700: #5c4ade · yellow-900: #ffd731 · green-200: #55db9c · green-700: #3e512f
Primary UI palette for PULSE landing: navy + light + blue-700 only (restrained). Accents only if earned.
```

Typography

- Primary: TWK Everett (Regular 400, Medium 500) — files in app/public/fonts/
- Mono: TWK Everett Mono Regular — labels, phase words, codes
- Tracking: large display titles ~ -0.05em; mono uppercase small labels with letter-spacing
- Font files already hosted: TWKEverett-Regular.otf, TWKEverett-Medium.otf, TWKEverettMono-Regular.otf

Chrome language

- 2px hard borders in navy (#000f1d) — no soft glassmorphism.
- Buttons: full-width navy bar + blue icon strip + arrow rotated -45deg (Overflow CTA language).
- Easing: cubic-bezier(0.525, 0, 0, 1).
- Background: light #f7f7f7 + blue grid SVG (public/grid-blue.svg) at controlled opacity/size.
- Selection: blue-700 background, navy text.

- Keycap-style elevation (face + side block) for iconic objects if used.
- Avoid rainbow key spam on landing; avoid empty minimalism; avoid fake “live preview” scoreboards that look mock.

Assets on disk

```
C:\Users\HP\pulse\app\public\fonts\*.otf  
C:\Users\HP\pulse\app\public\grid-blue.svg  
C:\Users\HP\pulse\app\public\grid-orange.svg  
C:\Users\HP\pulse\app\src\theme.css (token file)  
Overflow source CSS (study): https://cdn.prod.website-files.com/67acdc4f394bcf4f3e3669b6/css/sui-overflow-2-staging-ea69dc13ced32828.webflow.shared.50db99ae7.css
```

2.7 Quality bar (team standard)

- Mobile-first portrait ~390×844; max content width ~430px.
- Must look premium enough to win a mobile-themed hackathon UI review.
- Demo will be recorded on a real phone — thumb reach, safe areas, readable type.
- If design is “ok” or “simple empty”, it fails. Must be fire + clean + fact-based.
- Self-verify on phone viewport before any deploy claim.
- Human lead final approval before GitHub push of “perfect” landing.

3. Virtual machine & deploy environment

All live previews for the team happen on the existing production VM (same stack used for RECALL).

SSH host	ubuntu@51.21.252.153
SSH key (Windows)	C:\Users\HP\Downloads\cashual.pem
Example SSH	ssh -i "C:\Users\HP\Downloads\cashual.pem" -o StrictHostKeyChecking=no ubuntu@51.21.252.153
Domain	https://agentr.online
Static sites root	/var/www/agentr-sites/
Nginx rule	location /sites/ { alias /var/www/agentr-sites/; }
PULSE deploy path	/var/www/agentr-sites/pulse/
Public URL	https://agentr.online/sites/pulse/
Vite base	base: '/sites/pulse/' in vite.config.ts (required for assets)
TLS	Let's Encrypt certs on agentr.online (nginx)
Process manager	PM2 present (recall-app :3020, supermemory-local, agentr-api, etc.) — PULSE landing is static files under sites/
Deploy method	npm run build locally → scp dist/* to /var/www/agentr-sites/pulse/ → chmod a+rX
Do not	Re-setup entire VM infra; do not break /sites/recall/ or other sites

3.1 Local workspace

Project root	C:\Users\HP\pulse
FE app	C:\Users\HP\pulse\app
Plan markdown	C:\Users\HP\pulse\PLAN.md
GitHub remote	https://github.com/daraijaola/pulse
Git identity (machine)	daraijaola (credentials in ~/.git-credentials)
Node	v24.x available on builder machine
Rust/Anchor	Not required for pure FE landing polish; required later for onchain program

3.2 Deploy commands (reference)

```
cd C:\Users\HP\pulse\app
npm install
npm run build
ssh -i C:\Users\HP\Downloads\cashual.pem ubuntu@51.21.252.153 "rm -rf /var/www/agentr-sites/pulse/*"
scp -i C:\Users\HP\Downloads\cashual.pem -r dist/* ubuntu@51.21.252.153:/var/www/agentr-sites/pulse/
ssh ... "chmod -R a+rX /var/www/agentr-sites/pulse"
```

Deploy policy for expert FE AI: Prefer local preview (npm run dev) + screenshots/self-critique first. Deploy to VM only after a refinement cycle. Never call first draft “done.” Hard-refresh / cache-bust when verifying live URL.

4. Current codebase state (as of brief)

- Vite React TS app scaffolded under app/.
- Overflow fonts + grids copied into public/.
- theme.css holds Overflow CSS variables.
- Multiple landing iterations already attempted — quality not yet accepted by team lead.
- Expert FE AI should be free to redesign landing from first principles using this brief, keeping tokens/fonts/paths; do not inherit weak prior UI if better craft requires rewrite.

- Onchain program not yet the FE priority; mock game flow allowed on non-landing screens until chain wiring phase.

5. Expert AI prompt pack (copy-paste)

Use these prompts with a specialist frontend / mobile UI model. Feed this entire PDF (or sections 1–4) as context. Start with Prompt A only. Do not skip to full app.

5.1 Prompt A — Landing page (START HERE)

You are a world-class mobile product designer AND senior frontend engineer. You are extremely strict. Average work is failure. “Clean but empty” is failure. Rainbow clutter is failure. One-shot deploys are failure.

Team: Small hackathon team shipping PULSE for Solana Blitz v6 (MagicBlock). Human lead is ruthless on UI quality and records the demo on a real phone. You work in their repo and deploy only to their VM when the landing is truly ready.

Mission (this session only): Ship a PERFECT mobile landing page for PULSE. Nothing else. No full game wiring. No wallet. No backend. Landing only.

Product facts (do not invent mechanics):

- PULSE = mobile reaction battle: wait for a fair pulse/GO signal → tap first → win the round.
- Fair GO via MagicBlock VRF (not Math.random).
- Live room/taps on MagicBlock Ephemeral Rollup; settle winner on Solana.
- Solo demo uses a ghost opponent later; landing must sell the fantasy now.

Visual system (mandatory study):

Open and study <https://overflow.sui.io/> carefully (type scale, hard 2px navy borders, light #f7f7f7 ground, navy #000f1d ink, blue #4da2ff accent, keycap language, CTA with blue icon rail + -45° arrow, grid texture). Also load tokens from app/src/theme.css and fonts in app/public/fonts (TWK Everett + Mono). Use the Overflow system as craft reference. NEVER put Overflow/Sui logos or the words “Overflow system” in the UI.

Craft bar:

- Portrait phone-first, safe areas, thumb CTA.
- Simple AND fire – editorial confidence, not a wireframe, not a toy dashboard.
- Designed hero motion that embodies wait → hold → TAP → win (cinematic timing, not equal interval spam).
- Prefer TWO product facts only if you show facts: (1) fair pulse/VRF (2) live ER → Solana settle.
- No fake multi-color key spam. No “live preview” mock scoreboard chrome unless it is truly beautiful and product-true.
- Temporary wordmark “PULSE” OK; final logo later.

Workspace:

- Local: C:\Users\HP\pulse\app
- Vite base must remain '/sites/pulse/' for production assets.
- VM: ubuntu@51.21.252.153 key C:\Users\HP\Downloads\cashual.pem
- Deploy target: /var/www/agentr-sites/pulse/ → <https://agentr.online/sites/pulse/>
- Do not break other /sites/* projects.

Process (non-negotiable):

- 1) Read this brief + current app landing code.
- 2) Redesign landing from first principles if needed (rewrite allowed).
- 3) Build. Self-critique against Overflow craft + product truth + phone demo.
- 4) Refine at least 2–3 hard passes (spacing, motion, type, hierarchy) before deploy.
- 5) Only then build + scp to VM. Hard-refresh verify.
- 6) If not perfect, iterate again. Do NOT declare done after first upload.
- 7) Do not push to GitHub until human lead says the landing is perfect.

Deliverable: A perfect working landing page at <https://agentr.online/sites/pulse/> that makes a judge want to play, clearly signals the game, and looks like a winning mobile Blitz entry.

Begin now. Landing only.

5.2 Prompt B — After landing is approved (Enter + app shell)

Continue as the same strict mobile designer/engineer. Landing is locked unless a bug appears.

Next lead-source: **Enter (create/join)** then **Lobby** with Overflow chrome (top bar + dock) consistent with landing tokens. Huge room code, clear CTAs, ghost opponent labeling. Still no chain required if mocked, but structure must match final product. Same VM deploy path. Same refine-until-perfect rule. Phone demo first.

5.3 Prompt C — Arena + Result (game FE)

Implement Arena and Result screens to competition quality. Phase UX: delegate → VRF → wait → GO → tap → settle. TAP target must be thumb-first and beautiful. Motion must feel like the landing Signal language, not a different product. Mock chain OK; label states truthfully. Then prepare for wallet/ER wiring in a later backend-focused session. Refine until human lead approval; deploy VM; no rush push.

5.4 Prompt D — QA / perfection loop (any screen)

You are a strict design QA lead. Open the live URL and the code. List defects in hierarchy, spacing, motion, contrast, phone safe areas, copy truth, and Overflow-token fidelity. Fix only defects. Re-deploy. Repeat until none remain. Be brutal. "Looks fine" is not a pass.

6. Handoff checklist

- ■ Expert AI received this PDF + repo access + SSH key path.
- ■ Overflow site studied; tokens/fonts loaded.
- ■ Landing only until approved.
- ■ ≥2 refinement passes before VM deploy.
- ■ Live URL verified on real phone after hard refresh.
- ■ Human lead sign-off before git push of landing milestone.
- ■ Then Enter → Lobby → Arena → Result, one screen at a time.
- ■ Chain/ER/VRF after FE path is beautiful and demoable.
- ■ Demo video ≤90s following product loop.
- ■ Submit on Luma before Sunday deadline with repo + live + video.

End of master brief. If any fact conflicts with a newer human instruction, human instruction wins — update this PDF and re-issue.